



Question 1: Multiple Choice Question

WSP:

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40

- In Address Resolution Protocol (ARP), a packet is encapsulated directly into a
 - Data link header
 - Network frame
 - Network station
 - Data link frame
- The proxy server reduces the load on the
 - Proxy server
 - Node server
 - Domain server
 - Original server
- SMTP is a
 - Networking protocol
 - Protocol used for transferring message between eMail user and Mail Server
 - Encryption standard
 - Protocol used for smart card message interchange
- Which of the following functionalities must be implemented by a transport protocol over and above the network protocol?
 - End to end connectivity
 - Packet delivery in the correct order
 - Detection of duplicate packets
 - Recovery from packet losses
- Which one of the following protocol is used to resolve IPv4 address and Internet address?
 - WAN
 - RMP
 - ARP
 - Logical layer
- The physical layer contains with
 - agencies to process delivery
 - agencies to delivery
 - aggregation to application delivery
 - agencies to the network
- Which of the following is not an application layer service?
 - Network virtual terminal
 - File transfer, access and management
 - Virtual service
 - Error control
- Which of the following is a component of residential telephone net?
 - A high-speed downstream channel
 - A medium-speed downstream channel
 - An ultra-high-speed downstream channel
 - A low-speed downstream channel
- The combination of _____ and _____ is often termed the local address or the local portion of the IP address.
 - Network number and host number
 - Network number and subnet number



- c) Subnet number and host number
d) Host number
10. Transmission of digital data between two or more computers is called?
a) Internet
b) Network
c) Internet
d) Data communication
11. An Internet Protocol Version (IPv4) datagram can encapsulate data from several higher-level protocols such as User Datagram Protocol (UDP), RARP, ICMP and
a) CDNI protocols
b) ATM protocols
c) MPLS protocols
d) TCP protocol
12. Types of the data transmission medium
I) Twisted wire
II) Coaxial cable
III) Fiber optic cable
IV) Radio waves
a) I, II, III, IV
b) I, II, III
c) I, II, III, IV
d) None of the above
13. How many collision domains are created when you segment a network with 42-port switch?
a) 1
b) 2
c) 5
d) 12
14. How many hosts are on the network 172.16.41.0/28?
a) 65,534
b) 16
c) 254
d) 14
15. Which of the following protocols are examples of TCP/IP transport layer protocols?
a) all internet
b) HTTP
c) FTP
d) UDP
16. Communication channel is shared by two the machines on the network is
a) uni-cast network
b) broadcast network
c) multicast network
d) none of the mentioned
17. Which of the following IP addresses can be used as (a) loop-back address?
a) 10.0.0.0
b) 127.0.0.1
c) 255.255.255.255
d) 0.255.255.255
18. In computer network, nodes are
a) the computer that routes the data
b) the computer that terminates the data
c) the computer that originates the data
d) all of the mentioned
19. The physical path between sender and receiver through which the message or data travels is
a) switch
b) transmission medium
c) client
d) server
20. What is a Basic Router?
I) A device used to connect different networks altogether and Routes traffic between different networks using IP address.
II) A connecting device that makes Computer networking enables a much faster and more efficient communication mode than earlier postal service. This includes email, video communication, video conferencing etc.
III) It uses intelligent decisions, also known as routing protocols, in order to find the best way to get a packet of information from one network to another.
IV) It can either work as an expansion card, USB device or built and attached directly to the motherboard.
a) all of the above
b) none of the above
c) I, II, III
d) I, II, IV
21. Communication in which transfer of data takes place in both the directions and for a time
a) Full duplex communication
b) Half duplex communication
c) simplex communication
d) none of the above
22. What are the Methods to move data through a network of links and switches?
a) Packet switching and Line switching
b) Circuit switching and Line switching
c) Line switching and bit switching
d) Packet switching and Circuit switching
23. Types of data being transferred in each connection are numbered by TCP. These are start with a

24. A full-duplex communication channel permits information to travel
 a) At timed intervals
☒ b) Both ways, but not at once
 c) One direction only
☒ d) Both ways at once
25. Each IP packet must contain
 a) Only source address
 b) Only destination address
 c) Source and Destination address
☒ d) Source or Destination address
26. Which protocol does DHCP use at the Transport layer?
 a) IP
 b) TCP
☒ c) UDP
 d) ARP
27. There are the following statements that are given below, which of them are correct ARP?
 i. The ARP protocol is used to get destination MAC address from the destination IP address.
 ii. The ARP protocol is used to get the destination IP address from the destination MAC address.
 iii. The ARP stands for Address Resolution Protocol.
 iv. The ARP protocol works on the same network as well as connected networks
 a) i and iii
 b) ii and iii
 c) i, ii and iii
☒ d) i, iii and iv
28. Most packet switches use this principle
 a) Stop and wait
☒ b) Store and forward
 c) Store and wait
 d) Stop and forward
29. The required resources for communication between end systems are reserved for the duration of the session between end systems in _____ method.
 a) Packet switching
☒ b) Circuit switching
 c) Line switching
 d) Frequency switching
30. A local telephone network is an example of a _____ network.
 a) Packet switched
☒ b) Circuit switched
 c) Bit switched
 d) Line switched
31. What is main purpose of using subnetting?
 a) It divides a network into network classes
 b) It can speed up the network
☒ c) It divides one large network into several smaller ones
 d) None of the above
32. What does a VLAN do?
 a) Acts as the fastest port to all servers.
 b) Provides multiple collision domains on one switch port.
☒ c) Breaks up broadcast domains in a layer 2 switch internetwork.
 d) Provides multiple broadcast domains within a single collision domain.
33. What will be the subnet mask for the following IP address 100.101.110.100/11?
 a) 255.128.0.0
☒ b) 255.224.0.0
 c) 255.192.0.0
 d) 255.32.0.0
34. Which of the following the valid host range for the subnet on which the IP address 192.168.168.188 255.255.255.192 resides?
 a) 192.168.168.129- 190
 b) 192.168.168.128- 193
☒ c) 192.168.168.129- 191
 d) 192.168.168.128- 192
35. What is the broadcast address of prefix 172.18.16.0/22?
 a) 172.18.25.255
 b) 172.18.19.255
 c) 172.18.30.255
☒ d) 172.18.23.255
36. If you would like to give IP address to 30 hosts from your IP prefix which subnet mask you use most effectively?
 a) 255.255.255.128
 b) 255.255.255.240
 c) 255.255.255.224
☒ d) 255.255.255.252
37. There are the following statements that are given below, which of them are correct advantages of VLAN?
 i. Using VLAN, we can manage devices easily in the network.
 ii. Using VLAN, we can reduce the size of broadcast domains.
 iii. Using VLAN, we can add an additional layer of security.
 a) i, ii and iii
 b) ii and iii
 c) i, ii and iii
☒ d) i, ii, iii



38. Which of the following types of VLAN connections are supported by Switch?
 a) Access Link b) Trunk Link c) A and B d) None of the above
39. There are the following statements that are given below, which of them are correct VLAN?
 i. VLAN is used to divide a physical network into multiple logical networks.
 ii. VLAN is used to break large broadcast domains into smaller broadcast domains.
 iii. VLAN stands for Virtual Local Area Network.
 iv. VLAN stands for Visual Local Area Network.
 a) i and iii b) ii and iii c) i, ii and iii d) i, ii and iv
40. What do you understand about the physical layout in network topologies?
 a) The physical layout that describes the physical placements of network devices.
 b) The physical layout describes the flow of data within the network.
 c) The physical layout that describes cables only used in the network.
 d) All the above

QUESTION2:

2.1 Suppose Host A sends two TCP segments back to back to Host B over a TCP connection. The first segment has sequence number 80; the second has sequence number 100. (2 MARKS)

a. How much data is in the first segment?

b. Suppose that the first segment is lost but the second segment arrives at B. In the response from B, what will be the acknowledgment number?

2.2 True or false? (3 MARKS)

a. Host A is sending Host B a large file over a TCP connection. Assume that the data has been buffered in the sender. Host B will not send acknowledgments to Host A because Host A has not received the data. ()

b. Suppose Host A sends one segment with sequence number 38 and 4 bytes of data. Host B receives the segment and sends back an acknowledgment number 42. ()

c. Suppose Host A is sending Host B a large file over a TCP connection. The receiver has a buffer of 1000 bytes. The size of the receiver's buffer cannot exceed the size of the receiver's window. ()

d. What is the difference between a router and link-layer switch?

24-D

25-C

26-A

27-D

28-B

29-B

30-B

31-C

32-C

33-D

34-A

35-C

36-C

37-C

38-?

39-C

40-A